

Understanding the Impact of Natural Farming on Agrarian Distress in
Andhra Pradesh, India

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in partial fulfillment of the requirements for completion of the
Evidence and Inquiry certificate and the
Polymathic Scholars honors program in the College of Natural Sciences at
The University of Texas at Austin

Spring 2023

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Acknowledgements

This project began in 2020 when the pandemic made it seem uncertain and impossible. However, I have met some amazing people along this journey who have supported and encouraged me to write this paper. The case study was done in collaboration with the Center for Sustainable Agriculture (CSA), which helped organize field visits and provided local agriculture data on crops and demographics. Early on in my capstone field planning, I decided to conduct fieldwork with farmers, which became the central point around which I built my research. Many CSA staff members helped coordinate my research efforts, notably Dr. GV Ramanjaneyulu and Divya Veluguri, who helped develop my project and coordinate the visit to Kadapa. I also worked with highly knowledgeable faculty at the University of Texas, who guided my efforts. My first thesis supervisor was Dr. Calvin Streeter. Dr. Streeter gave me the initial support I needed to see my project as a timeline rather than an impossible task. He helped define my project and figure out the logistics of my pilot study. As Dr. Streeter retired, he introduced me to Dr. Shiv Ganesh. I shared with Dr. Ganesh that I felt removed from discussing agrarian distress and agriculture when I was not in India. He then introduced me to Dr. Sharmila Rudrappa and the wonderful President Award For Global Learning student team. The South Asia Institute helped partially fund my trip to India. I also want to thank Dr. Rebecca Wilcox and Dr. Sara Corson. This paper and research would not have been possible without these people and the Polymathic Scholars Honors Program. Lastly, I want to thank the farmers in Kadapa that welcomed me with open arms and allowed me to become a part of their lives while I was doing my field visits.

Abstract

Sustainable agriculture is highly esteemed as a worldwide solution to environmental and food security concerns. Specifically in India, sustainable agriculture practices rehabilitate the land and create a sustainable livelihood for farmers facing increased debt and crop loss. Many regional sustainable practices have gained momentum recently, such as Andhra Pradesh Community-Based Natural Farming (APCNF) which originated from the Zero Budget Natural Farming program (ZBNF). These programs address the economic, ecologic, and social stressors which cause agrarian distress. The APCNF program works to help conventional (chemical) farmers learn about and shift to natural farming practices, which promotes reduced chemical inputs. Through a narrative literature review, this project provides a detailed background on agrarian distress and sustainable farming in India while discussing ZBNF and APCNF in depth. This background is followed by a case study of Kadapa, Andhra Pradesh (AP) which contextualizes the factors that support a farmer's transition to natural farming through a thematic analysis of the causes of the transition from conventional to natural farming. The paper specifically focuses on small, marginal farmers, who are the most vulnerable to climate change variations and market price fluctuation. Preliminary findings show that farmers are transitioning to natural farming due to three key factors: (1) CSA and community support, (2) the impact of chemicals on farming practices, and (3) the improvement of both quantity and quality of produce. These factors are interlinked and help farmers create sustainable livelihoods. Natural farming practices address the root causes of agrarian distress and should be further studied and implemented across India.

Key terms: sustainable agriculture, climate change, natural farming, Andhra Pradesh Community-Based natural farming (APCNF), Zero Budget Natural Farming (ZBNF)

Introduction

Agrarian distress is defined as the challenges farmers face when their agricultural livelihood becomes financially unsustainable due to socioeconomic conditions (Southard & Randell, 2022). Agrarian distress is a common issue farmers face in South Asian countries, characterized by the financial burden that follows crop failure. Unfortunately, this has led to a significant number of farmers who die by suicide in the region. Though this issue is prevalent worldwide, India, in particular, has had an increasing number of farmers who die by suicide due to agrarian distress. A study conducted by Nagaraj et al. (2014) from 1997-2012 found that an estimated 264,388 farmers had died by suicide in India during that time. This statistic points to a concerning issue in India as a significant portion (45%) of the population depends on agricultural activities (Ministry of Agriculture & Farmers' Welfare, 2020). Small and marginal farmers of the population dependent on agriculture hold 85% of all landholdings (Ministry of Agriculture & Farmers' Welfare, 2020). These small marginal farmers are the most vulnerable to financial instability and least resilient to climate change impacts, putting them at higher risk of agrarian distress (Morton, 2007; Sahay, 2022).

This paper aims to explore the relationship between agrarian distress and sustainable agriculture, which is important as it indicates and evaluates solutions to agrarian distress. To achieve this goal, the study utilized a two-part methodology: a literature review and a case study. The literature review focused on agrarian distress and sustainable agriculture, with a particular emphasis on articles published in the past ten years; however, some older sources were included. Indian government reports were also consulted to obtain relevant agricultural data. Following the literature review, a case study was conducted in collaboration with the Center for Sustainable

Agriculture in India. The study was conducted in Kadapa, Andhra Pradesh, India, and involved semi-structured interviews with natural farmers. The data collected from the interviews were analyzed using a thematic analysis approach to identify key themes and patterns in the farmers' experiences with sustainable agriculture and agrarian distress. Overall, this study aimed to provide a comprehensive understanding of the current state of agrarian distress and sustainable agriculture in India and to highlight the perspectives and experiences of natural farmers in the region.

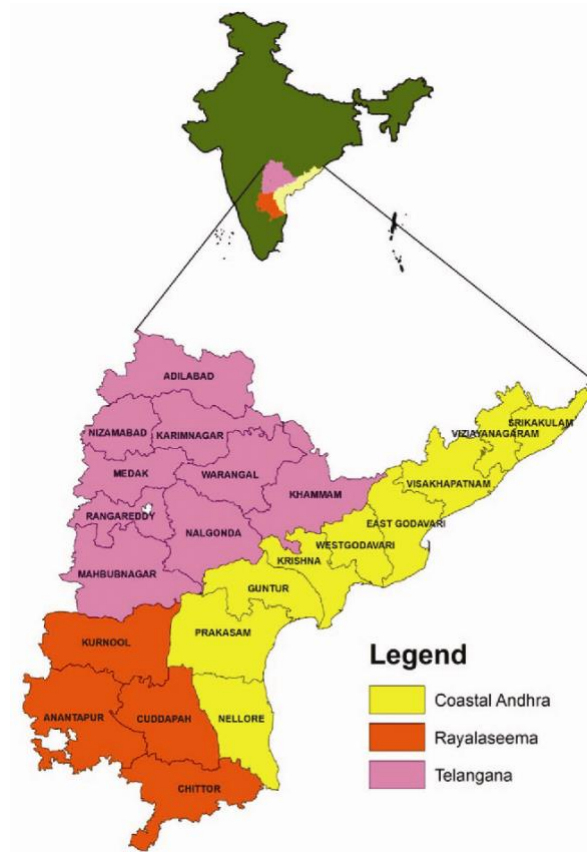
Historically, the Green Revolution in India introduced chemicals and high-yielding crops in a top-down approach controlled by large corporations (Parayil, 1992). However, this study focuses on the critical shift toward community-based systems, such as the Andhra Pradesh Community Managed Natural Farming program, which emphasizes farmer collaboration and support (Galab et al., 2022; Jaacks et al., 2023). This paper draws on fieldwork conducted in the Kadapa district of Andhra Pradesh to highlight why farmers are transitioning to and sustaining natural farming. Including the Kadapa case study in this paper elevates the role of farmers as front-line workers in the sustainable agriculture movement. This case study presents a critical perspective that re-centers the farmer at the heart of discussions on sustainable agriculture.

Given the critical role of agriculture in providing food security for society, it is imperative to have a meaningful discussion on agrarian distress in regards to farmer welfare. The hope is that this research and future research on sustainable agriculture will inspire the development of similar programs that aim to alleviate the burden of this issue and promote programs that benefit both farmers and society.

Agriculture in Andhra Pradesh (AP) State and Kadapa District

Figure 1

Map of India with Andhra Pradesh and Telangana Highlighted



Note. Andhra Pradesh is located in the southeast corner of India. The location map lists districts in Andhra Pradesh and Telangana. The case study in this paper takes place in Kadapa (also spelled Cuddapah) which is located in the in the Rayalaseems area. From “Climate Change Analysis in Southern Telangana Region, Andhra Pradesh Using LARS-WG model,” by Reddy, K. S., Kumar, M., Maruthi, V., Umesha, B., Vijayalaxmi, & Rao, C. V. K. N, 2014, *Current Science*, 107(1), p. 54–62. (<http://www.jstor.org/stable/24103404>). Copyright 2014 by Central Research Institute For Dryland Agriculture.

In Andhra Pradesh (AP), 60% of the population depends on agriculture, and this sector contributes to 19% of the state's GDP (Reddy, 2011). The climate of AP includes two main seasons - Rabi and Kharif. The Rabi season is from October to March and is characterized by cool and dry weather, which is favorable for cultivating crops such as wheat, barley, and pulses (Basava Raju et al., 2021). The Kharif season starts in June and lasts until October and is characterized by high humidity and heavy rainfall, which is favorable for cultivating crops such as rice, maize, cotton, and sugarcane (Basava Raju et al., 2021). However, the region is also vulnerable to the effects of climate change, which is increasingly affecting agricultural productivity and exacerbating agrarian distress faced by farmers.

The climate and topography of Andhra Pradesh are conducive to cultivating various crops, and the state is one of the largest rice producers in India (Basava Raju et al., 2021). Other crops such as cotton, sugarcane, tobacco, chili pepper, groundnut, mango, sunflower, and other horticulture crops are also cultivated (Basava Raju et al., 2021). Despite the favorable climate and abundant natural resources, farmers in Andhra Pradesh faces several challenges, such as water scarcity and soil degradation, which in turn cause low productivity. Furthermore, crop cultivation heavily depend on the climate, and due to climate change, there has been an increase in extreme weather, such as drought, cyclones, and river floods all which cause crop loss (Lange et al., 2020). Overall, climate change has caused a shift in cropping patterns and cropping choices. To mitigate the crisis and enhance farmers' livelihoods, the Andhra Pradesh government pushed the statewide effort of APCNF to convert 8 million acres and 6.5 million farmers to natural farming practices by 2024 (Rosenstock et al., 2020).

Figure 2

Map of Kadapa District in the state of Andhra Pradesh



Note. There are various spellings to the district name and both Kadapa and Cuddapah are in reference to the same region. From “Kadapa district map,” by Mapience India Limited, 2001, Directorate of Medical Education (<https://dme.ap.nic.in/nodaldme/dist/Cuddapah.html>).

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Kadapa is one of the 13 districts in Andhra Pradesh, located in the southern part of the state. The district is known for its diverse agricultural practices, including cultivating paddy, groundnut, sugarcane, and cotton (Government of Andhra Pradesh, n.d.). The region has a semi-arid climate with an average annual rainfall of around 700 mm (Badri et al., 2019). This rainfall is highly variable and often inadequate for proper crop production and leading to chronic droughts and crop failure (Basava Raju et al., 2021). These events have a significant impact on the region's agricultural sector and the livelihoods of local farmers. The major cause of desertification in Kadapa is the increasing pressure on land due to population growth and

urbanization, leading to the expansion of agricultural land and the overuse of natural resources such as water and soil (Bhaskar et al., 2021). Overgrazing and unsustainable agricultural practices, such as monoculture farming and excessive use of chemical fertilizers and pesticides, have also contributed to soil erosion and land degradation (Bhaskar et al., 2021). The impacts of desertification in Kadapa are severe and include soil erosion, loss of soil fertility, reduced agricultural productivity, and loss of biodiversity (Bhaskar et al., 2021). These ecologic factors have led to increased indebtedness and migration of farmers, exacerbating agrarian distress in the region.

This background sets the stage for the following conversation on agrarian distress and the response of Zero-budget Natural Farming (ZBNF) and Andhra Pradesh Community-based Natural Farming (APCNF).

Agrarian Distress

Agrarian distress is caused by the numerous interlinked stress factors farmers face. As the population of India grows, the demand for food increases which puts a pressure on the food system to meet the population needs. Along with the increased demand, there has been a decrease in yield due to many reasons, including, but not limited to, shifting climate patterns, groundwater depletion, soil quality degradation, and the increased price of chemical inputs (Jain et al., 2015; Meena et al., 2020; Reddy et al., 2014).

For farmers, creating a sustainable livelihood has become difficult as the land is exhausted. Adding more chemicals under the assumption that it will nourish the soil backfires, further degrades soil quality, and reduces the yield. Before the cropping seasons, farmers initially take out loans and their savings to finance their expenses, including seed, labor, technology, and

chemical inputs. When the farmer doesn't make enough profit to repay the loan, it results in them taking out another loan to begin the cropping season. In this manner, farmers get stuck in a vicious cycle of indebtedness (Maurya & Vishwakarma, 2021). Indebtedness is one of the main factors causing farmer to die by suicide (Merriott, 2016). Annually, 16,500 farmers die by suicide which is about a 50% higher rate of suicide than the general population (Merriott, 2016).

Ecological Stress Factors

Agrarian distress can be affected by several ecologic factors that reduce the land's capacity and decrease the farmer's profit. Many farmers in India are small, marginal farmers who are at risk of financial crisis due to profit loss, and they depend on rain-fed crops which are sensitive to variations in precipitation (Kumar & Parikh, 2001; Morton, 2007). Weather variability is a challenge to farmers as their cropping cycle is linked to the climate, and if there are unexpected variations in precipitation and temperature patterns, this can cause crop failure (Kumar & Parikh, 2001).

Jain et al. (2015) conducted a study on the causes and consequences of delayed monsoons in Gujrat, India, found that farmers have adapted to switch their crops to account for the variation. The adaptations of the farmers depend on their source of irrigation: rain-fed, canal, or groundwater. It also depends on the resources available to the farmer. Wealthier farmers have an easier time adapting to the monsoon delay by irrigating their crops more to maintain profit, but poorer farmers must maintain their profit by switching their crops to a more resilient crop, such as castor. Jain et al. (2015) also found that farmers may also delay their sowing dates to adapt to the delayed monsoon.

There is a similar effect on farmer cropping decisions with natural disasters, which have been exacerbated and frequent due to climate change. In the Rayalaseema region of Andhra

Pradesh (Figure 1), there has been monsoon failure with reduced rainfall which has caused chronic drought in the area (Muthana & Manne, 2018). The percentage of irrigated lands in the region has continuously decreased, which has caused a shift away from cultivating food crops, which are more resource intensive, to non-food commercial crops. Water scarcity causes crop failure and decreased water availability for human consumption and livestock sustenance (Muthana & Manne, 2018).

Reliable water availability is integral to the cultivation of healthy crops and has a tremendous impact on farming activities and decisions made by farmers. Beyond delayed monsoon and reduced rainfall, farmers are also struggling with the crisis of ground water depletion. The Central Ground Water Board of India defines critically depleted regions as areas with long-term reductions in groundwater levels (Singh et al., 2022). Irrigation for agriculture in India is primarily dependent on groundwater; however, the over-extracting from wells has caused groundwater depletion (Jain et al., 2021; Rodell et al., 2009). According to Jain et al., (2021) cropping intensity will decrease by an estimated 20% in India and by 68% in critically depleted regions with the highest level of stress to be faced by 2025. Decreased cropping intensity indicates reduced profitability of agriculture, which further perpetuates agrarian distress in regions with severe water scarcity.

Soil degradation and erosion is another ecological concern that dramatically decreases agricultural productivity. Agrochemicals such as pesticides and fertilizers can kill beneficial soil microorganisms and pests that would otherwise increase soil quality and help sustain healthy crops (Meena et al., 2020; Nayak & Solanki, 2021). The overuse of chemicals can cause increased soil acidity, which makes it difficult for proper nutrient uptake from the soil (Meena et al., 2020). When crops cannot properly source their nutrients, the overall resilience of the plant

decreases making it susceptible to pests. Furthermore, human activities such as deforestation and unsustainable farming practices such as overgrazing are perpetuating the issue of soil erosion on fields (Periasamy & Shanmugam, 2022). Soil health is vital for agricultural productivity, and the use of agrochemicals causes soil degradation, which reduces crop yield and ultimately puts financial pressure on the farmer which, leads to agrarian distress.

Economic Stress Factors

The ecologic stress factors are linked to crop loss which ultimately become a financial burden on the farmer. Furthermore, farmers struggle with direct economic stress factors regarding fluctuating crop prices and inadequate access to credit (Merriott, 2016). Before the sowing season, the Department of Agriculture and Co-operation government of India government sets the minimum support price (MSP) for certain crops. However, while MSP keeps market price of rice and wheat relatively stable, the prices offered to farmers fluctuate based on supply and demand for other crops. For example, the rates of common ingredients of onions and tomatoes heavily fluctuate as production is seasonal and high production leads to a fall in prices. If a crop is overproduced, then there is decreased demand and lower prices for the product which means the farmer may not meet their expected income. For crops that are not perishable, like cotton, farmers can hold till a better price is available at the market for their goods.

Once the harvest season concludes, farmers begin planning for the next sowing season and require money to reinvest in their fields. Farmers typically seek this investment through credit, however, farmers struggle with inadequate access to credit, typically used to purchase seeds, fertilizer, and agro-machinery, among other expenses (Rosenstock, 2020). Farmers need to take out loans to invest in their land due to limited financial literacy and lack of collateral

(Gautam et al., 2022). The necessity of investment and lack of credit access leads farmers to depend on private moneylenders with high-interest rates to finance their cropping seasons (Sadanandan, 2014). The leading cause of farmer suicides in India is the increasing debt burden on the farmer. Economic factors are directly linked to the resources available to the farmer, and decreased profit and increased debt from economic challenges exacerbate the effects of agrarian distress.

Social Stress Factors

Along with economic and ecological factors, social factors also affect agrarian distress. Small and marginal farmers have a large portion of all landholdings in India, and these farmers have a difficult time financially recovering losses of crop failure due to the ecological factors listed above (Sahay, 2022). Low-income farmers have limited access to resources; thus, it becomes difficult for farmers to improve their living standards. A 2019 UN report on the Food and Agriculture Organization's (FAO) work regarding climate change found that small marginal farmers are disproportionately impacted by climate change. Small marginal farmers do not have the resources to recover from multiple extreme weather events and the loss of agricultural productivity.

Farmers are mainly turning to income diversification which reduces their reliance on agriculture. Diversification can include increased reliance on livestock, and it could also be labor migration which causes a livelihood shift away from agriculture (Burnham & Ma, 2015). This migration is linked to the feminization of agriculture as more men move to cities to work, and women are left to make decisions and work in the fields (Pattnaik et al., 2017; Southard & Randell, 2022). However, it is essential to note that these women do not hold land titles. As more women enter the fields, they take on these responsibilities and household duties, which add more

stress to their lives (Pattnaik et al., 2017). Land ownership makes farmers eligible for some monetary government schemes. However, many tribal and female farmers do not have documentation for the land they work on and thus do not receive government aid. Land ownership and leasing are also an issue as many farmers in India are tenant farmers, and they are under pressure to show a profit while also working with the demands of the market.

In summary, agrarian distress is a complex issue that arises from the interplay of ecological, economic, and social factors. Crop yield heavily relies on natural resources, and any disruption in rainfall or soil quality can lead to a poor growing season, adversely affecting a farmer's profitability. Farmers are further disadvantaged by fluctuating market prices and limited access to credit, making it difficult for them to achieve financial stability. Small and marginal farmers are especially vulnerable to the impacts of climate change, as they lack the resources to sustain multiple crop failures. In addition to economic consequences, agrarian distress causes labor migration, which carries various social implications, disproportionately affecting women who are often at the center of agricultural functions. These factors are not siloed but are heavily interlinked, directly and indirectly impacting each other.

To address this issue, govt of AP promoted the APCNF program, and also partnered with various non-profits to implement the same program. This initiative leverages the existing rural community structure to help farmers reduce their reliance on chemical inputs and adopt more sustainable farming practices.

Sustainable Agriculture: Zero Budget Natural Farming (ZBNF) and Andhra Pradesh Community-Based Farming (APCNF)

Sustainable agriculture is a type of farming that prioritizes the balance between food production and natural resource conservation (Duddigan et al., 2022). Sustainable agriculture practices stand in contrast to conventional agriculture practices, which heavily depend on chemical fertilizers and pesticide use. In India, a sustainable agriculture program called Zero Budget Natural Farming (ZBNF) aims to address the ecological stress factors discussed in the previous section through a spiritual and holistic perspective. Andhra Pradesh has taken inspiration from this program and created a state-wide program called Andhra Pradesh Community-Based Natural Farming to further address the social and economic stress factors that cause agrarian distress. The details of both of these programs along with their relation to addressing agrarian distress are discussed in this section.

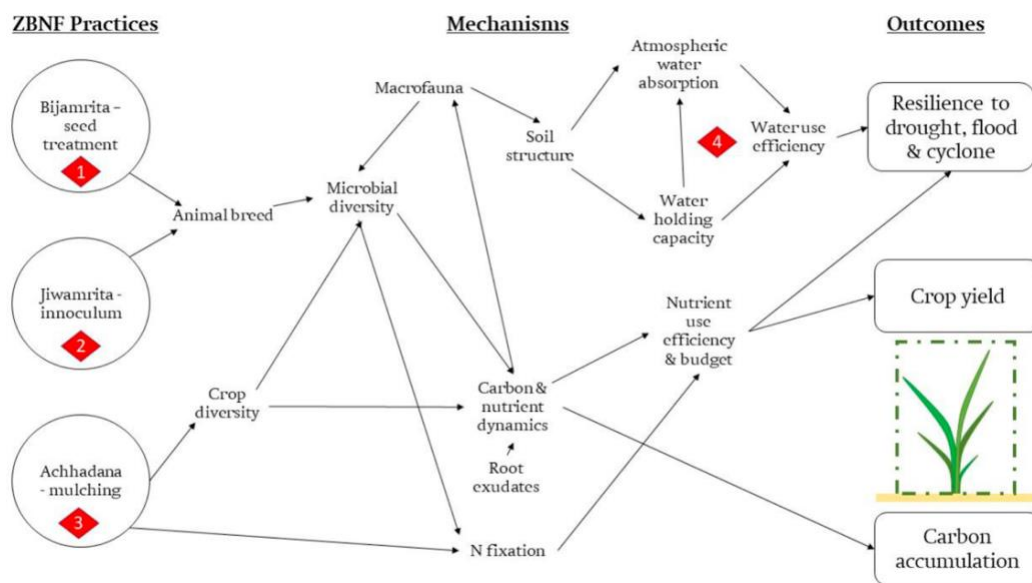
Zero Budget Natural Farming (ZBNF)

Zero Budget Natural Farming (ZBNF) is a farming technique developed by the Indian agriculturalist Subhash Palekar. The goal of the technique is to create self-sufficient and low-cost agriculture practices that follow along with the natural processes that take place on the fields (Khadse et al., 2017). Natural farming uses inputs such as cow urine, cow paddy, fish amino to source naturally occurring nutrients that help crops grow. The practice also encourages farmers to use indigenous seeds and avoid monoculture farming to help revitalize the land (Khadse et al., 2017). Though the technique does aim to reduce the cost incurred by the farmer the term "zero-budget" is misleading because it implies that the farmer has no expense. However, there is some cost associated with the technique as farmers need seeds, access to cattle and require agricultural

machinery to cultivate their crop. However, since the farmer uses easily accessible or readily available natural products, the general cost of cultivation is reduced (Khadse et al., 2017).

Figure 3

Flow chart showing field applications of the four principle ZBNF practices



Note. The flow chart contextualizes the practical implications of the four ZBNF pillars. From “Impact of Zero Budget Natural Farming on Crop Yields in Andhra Pradesh, SE India,” by Duddigan, S., Collins, C. D., Hussain, Z., Osbahr, H., Shaw, L. J., Sinclair, F., Sizmur, T., Thallam, V., & Ann Winowiecki, L, 2022, *Sustainability*, 14(3), p. 1689. (<https://doi.org/10.3390/su14031689>). Copyright 2022 by authors.

The four central pillars of ZBNF, as shown by Figure 3, are *bijamrita*, *jiwamrita*, *achhadana*, and *whapasa* (Khadse et al., 2017; Tiji, 2023). The appeal of the ZBNF practice is the low-cost nature which is evident through the “recipes” used to make natural fertilizers and pesticides. To create *bijamrita*, farmers prepare a cow urine and dung mixture with animal byproducts, water, and lime to protect young seed roots from fungus and make the seed resistant to illnesses that cause rot during the monsoon season. To create *jiwamrita*, farmers used cow

urine, cow dung, jaggery, live soil, gram flour, and water mixture used as fertilizer for crops and sprayed on crops to increase yield. *Achhadana*, the third pillar, is the practice of mulching and leaving past crop remnants as ground cover for the next crop to recycle nutrients (Khadse et al., 2017). Since the crops are remnants of the past season, the farmer is able to reutilize them as mulch to help reduce water evaporation and maintains soil moisture. *Whapasa* is a water management method to efficiently water crops with minimum wastage by managing watering times (Khadse et al., 2017). Palekar claims that reducing irrigation to only be done at noon helps plants utilize the water vapor and air molecules to soften the soil (Khadse et al., 2017). These pillars not only aim to reduce cost, but also promote proper natural resources management to improve the ecological impact of agriculture.

Building off of the ZBNF program and its practices, Andhra Pradesh has adopted a variation of the program for its farmers. This program is called Andhra Pradesh Community-Based Natural Farming also known as Andhra Pradesh Community-Managed Natural Farming (APCNF).

Andhra Pradesh Community-Based Natural Farming (APCNF)

APCNF is an initiative by the Andhra Pradesh Department of Agriculture which is promoting the program through the nonprofit organization of Rythu Sadhikara Samstha (Bharucha et al., 2020; Duddigan et al., 2022). This program builds on the ZBNF practice and aims to promote natural farming through community empowerment and participation (Khadse & Rosset, 2019). Under this model, groups of small farmers form a community-based organization that works together to develop and implement sustainable farming practices. These practices may include using natural fertilizers and pesticides, crop diversification, conservation of natural resources, and adopting modern technologies (Jaacks et al., 2023). Community support creates

an environment of knowledge and resource sharing to help farmers who are unfamiliar with natural farming practices.

The APCNF model improves small farmers' income and livelihoods by promoting sustainable agriculture practices and empowering communities to take charge of agricultural development (Jaacks et al., 2023). The government of Andhra Pradesh has been actively promoting this model as part of its efforts to encourage sustainable agriculture and improve the economic conditions of farmers in the state. Overall, the ZBNF practices have shown to decrease expenditure and increase yield which, dependent on market price, equates to increased profit margins for the farmers who are most vulnerable to agrarian distress (Bharucha et al., 2020). However, there are many factors that impact the result of natural farming practices such as crop type and agroclimatic conditions with more precipitation (Duddigan et al., 2022; Ponisio et al., 2014).

Though the APCNF program is based on ZBNF practices, it is important to distinguish ZBNF program from the APCNF program promoted in the state of AP. According to Veluguri (2021), the APCNF program uses the base knowledge put forth by Palekar but goes beyond the principles outlined and works to include more chemical-free practices. An example of this distinction would be the use of vermicompost. The APCNF program promotes using non-native earthworks to promote soil health which contradicts Palekars' teachings (Veluguri, 2021). The APCNF program has built off the ZBNF four pillars and follows nine principles. The principles are listed below (Galab et al., 2022).

1. Soil to be covered with crops 365 days (living root)
2. Minimal disturbance of soil
3. Bio stimulants as necessary catalysts

4. Use indigenous seed
5. Diverse crop, trees 15-20 crops
6. Integrate animals in to farming
7. Increase organic residues on the soil
8. Pest management through botanical extracts
9. No synthetic fertilizers, pesticides, and herbicides

The general background on the functions and goals of the ZBNF and APCNF programs provides context for how the programs are benefiting farmers. The following section exploring the ecological, economic, and social factors linked to APCNF and ZBNF, is organized parallel to the section on agrarian distress to illustrate how the programs address the main stressors causing agrarian distress.

Ecological Benefits

Ecological restoration is at the forefront of natural farming and sustainable agriculture efforts. Starting from the ground up, natural farming methods improve the soil twofold. First, natural farming promotes reduced use of chemical fertilizers which decreases the harm caused by chemical runoff (Bharucha et al., 2020). This can look like replacing chemical pesticide with non-pesticidal management practices (NPM) where the farmer uses pheromone traps and trap crops to attract pests and promote beneficial insects to establish a healthy biome (Bharucha et al., 2020; Khadse & Rosset, 2019). Second, farmers work to recover their damaged soil with natural inputs such as manure and fish amino (Krishnamoorthy et al., 2018; Tiji, 2023). The inputs farmers use is dependent on the cropping cycle. At the beginning of the cropping season, farmers till manure into their soil to soften the soil and add nutrients to help the seeds and seedlings grow with well-established root systems. Fish amino is made with fermented fish carcass from local

fish markets and helps boost soil fertility and softens soil to create a conducive environment for beneficial soil organisms such as earthworms (Bharucha et al., 2020; Krishnamoorthy et al., 2018; Tiji, 2023). The ecologic advantages of APCNF and ZBNF are also integrated with the economic advantages as it promotes the use of low-budget inputs and integrated farming practices (Bharucha et al., 2020).

Economic Benefits

ZBNF is not truly “zero-budget” but requires significantly lower external input costs which set the farmer up to increase their profit margin when they sell their harvest (Moses Shyam et al., 2019). The increased profit margin is because ZBNF promotes the use of locally available resources and advises farmers to produce inputs themselves to decrease costs. For example, the fish amino concoction is made from fish waste and is typically sold at a nominal cost or collected for free from local markets. Another way farmers increase their profit margin is by cultivating low resource intensive native crops, which have ecological benefits of their own. Farmers typically leave the land fallow between cropping cycles to recover and prepare for the next crop. At this time, natural farming practices promote the growth of native varieties of crops such as millet. In AP, *sajjalu* (millets) and *jonnalu* (Sorghum) are *naatu* (native) varieties of crops in the region that are grown in the fallow lands (Bharucha et al., 2020). The crops require less water and maintenance than the regular crop, and at the end of the season, farmers can harvest and consume the highly nutritious crop or feed the crop to poultry and cattle as fodder (Bharucha et al., 2020). These multipurpose native seeds also have ecological benefits. Millets and sorghum are also considered high-residue crops that protect the topsoil from blowing away or eroding in runoff (Bharucha et al., 2020; Coulibaly et al., 2020). Of the crop that is remaining

at the end of the season, farmers can leave as mulch for the following crop (Bharucha et al., 2020).

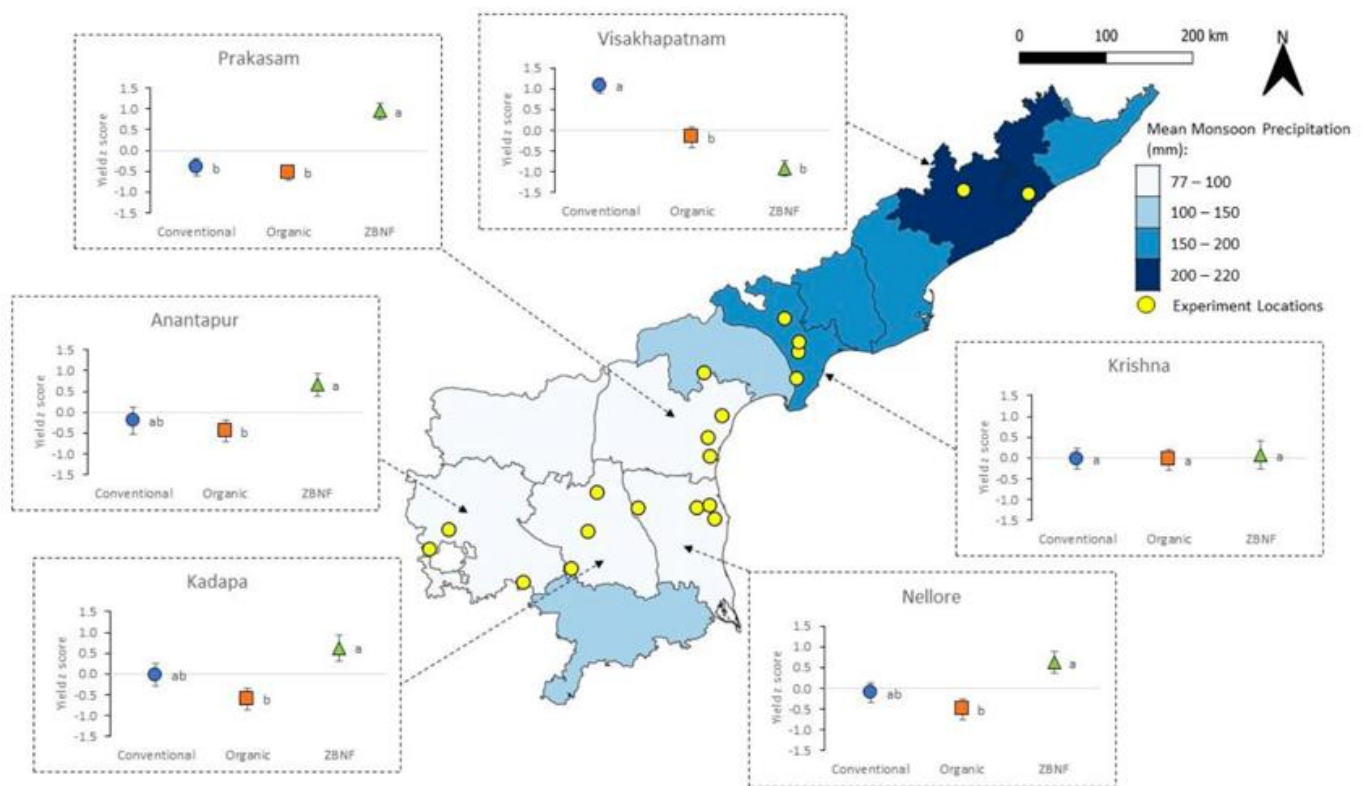
Beyond conscious natural farming practices, research (Duddigan et al., 2022) shows that the yield is also increased by ZBNF practices in AP, India. A 2022 study conducted by Duddigan et al. evaluated yield from various fields across AP, comparing conventional (chemical), organic, and natural farming (Figure 4). The study was done in the Kharif season with twenty fields across six districts across the states, and each fields had a variety of horticulture crops. The large area of study accounts for climate variations in the region which makes the results more credible. The method of evaluation for the study included applying the three farming practices to various plots of land on the same field. The yield was calculated per plot in Andhra Pradesh during the 2018 growing season and found that, on average, across the districts, ZBNF farms were able to produce more yield than organic and conventional farming (Figure 5).

The results of the study showed that there would not be any crop loss for farmers who plan to shift to natural farming (Duddigan, 2022). This finding disputes the common criticism of natural farming that farmers increase their risk by incur losses when transitioning to natural farming (Duddigan, 2022). Since the data collected was for a single growing season, it is credible in showing the short-term impacts of the ZBNF practice. Furthermore, the district wise analysis shows that the monsoon precipitation differences vary across the state, and Figure 4 shows that districts that receive adequate rainfall have an inverse relationship with yield from the ZBNF program. Southern, more arid districts have increased output from ZBNF practices. Studying the northern districts shows that increased precipitation has a reverse impact on the yield from ZBNF practices.

Though the short-term comparative results of the ZBNF program show an increased yield, the long-term impact and climate relationship of the ZBNF practices in terms of increased yield over multiple growing cycles needs to be studied and evaluated.

Figure 4

Map of Andhra Pradesh, relating the yield outcomes from conventional, organic, and ZBNF practices across districts

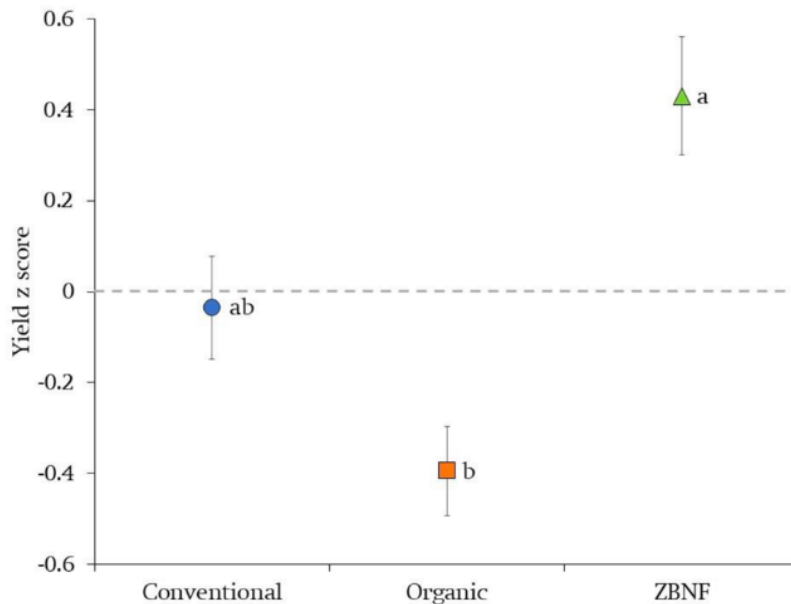


Note. The map shows the locations of the twenty experimental fields and the mean monsoon precipitation of each district. The mean yield graph of each selected district are shown to the side of the map. From “Impact of Zero Budget Natural Farming on Crop Yields in Andhra Pradesh, SE India,” by Duddigan, S., Collins, C. D., Hussain, Z., Osbahr, H., Shaw, L. J., Sinclair, F., Sizmur, T., Thallam, V., & Ann Winowiecki, L, 2022, *Sustainability*, 14(3), p. 1689.

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Figure 5

Mean results from the twenty experimental fields of 2018 study comparing the yield of the three farming practices (conventional, organic, ZBNF)



Note. Though there are variation in the yield across the districts based on climate conditions, the mean yield data does show a higher yield for the plots that used ZBNF practices. From “Impact of Zero Budget Natural Farming on Crop Yields in Andhra Pradesh, SE India,” by Duddigan, S., Collins, C. D., Hussain, Z., Osbahr, H., Shaw, L. J., Sinclair, F., Sizmur, T., Thallam, V., & Ann Winowiecki, L, 2022, *Sustainability*, 14(3), p. 1689.

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Social Benefits

As stated in the agrarian distress section on social stressors, the main populations that are vulnerable to agrarian distress are small farmers and female farmers who cannot cope with crop loss. To address this issue the APCNF program targets small, marginal farmers and female farmers to become a part of the program (Khadse & Rosset, 2019). Smallholder farmers in

Andhra Pradesh face several challenges, including low productivity, high production costs, and the adverse impacts of climate change. The use of chemical fertilizers and pesticides is one of the major factors contributing to these challenges, as it reduces soil fertility, harms beneficial soil organisms, and poses health risks to farmers and consumers (Nayak & Solanki, 2021). These farmers who do not have a margin to try new methods are typically stuck in a cycle of losses (Merriott, 2016).

The APCNF program provides training and support to smallholder farmers to adopt natural farming practices such as composting, mulching, intercropping, and the use of natural fertilizers and pesticides. The program also promotes the cultivation of traditional native crops and encourages farmers to form farmer-producer organizations (FPOs) to market their produce and improve their bargaining power collectively (Khadse & Rosset, 2019). The impact of the APCNF program on smallholder farmers has been positive. Farmers who have adopted natural farming practices have reported increased yields, improved soil health, and reduced production costs (Duddigan et al., 2022; Laishram et al., 2022).

The APCNF program has empowered smallholder farmers, particularly women, by providing them with training and support to adopt sustainable farming practices (Paliath, 2022). The program has contributed to biodiversity conservation by promoting local, traditional crop cultivation and preserving indigenous seeds. Overall, the APCNF program has significantly impacted smallholder farmers in Andhra Pradesh by promoting sustainable agriculture practices, improving livelihoods, and enhancing the resilience of farmers to the adverse impacts of climate change.

In summary, the ZBNF and APCNF programs aim to address the ecological, economic, and social factors that cause agrarian distress. While the program cannot solve climate and yield

variations, it helps farmers better adapt to the changes. Though the Duddigan et al. (2022) study did show a short-term increase in yield with ZBNF practices, the transition away from conventional farming still requires farmers to take a risk on their fields. The farmers' voice is critical in this transition, as it explains their concerns and motivations while outlining how community support can help fill any apprehensions about natural farming. Organizations such as the Center for Sustainable Agriculture (CSA) have collaborated with APCNF to help implement the program at a more targeted capacity in Andhra Pradesh.

Case Study: Kadapa, AP, India

Methods

In the winter of 2021, I had the opportunity to conduct a pilot study in Ananthapur, Andhra Pradesh, India with Verdentum, an impact analysis organization. During this study, I joined the Verdentum team on a visit to Accion Fraterna (AF) Ecology in Ananthapur to gain insight into agrarian distress and the current programs available to support farming families. While there, I spoke with field workers who were actively promoting natural farming in the region, and it was during these conversations that I first learned about APCNF and ZBNF.

Through my discussions with staff members at AF Ecology, I was able to connect with a larger network of individuals who were working to promote natural farming, including Dr. Ramanjaneyullu and Divya Veluguri from the Center of Sustainable Agriculture (CSA). After expressing my desire to conduct fieldwork to gain a deeper understanding of the farmers' perspective on natural farming, they assisted me in developing and translating a semi-structured interview that focused on this topic.

In the Summer of 2022, field visits in Kadapa, AP were done in collaboration with CSA. CSA is an independent research organization based in Hyderabad, AP, India that collaborates with non-profit organizations and government agencies across the Indian states of Andhra Pradesh, Maharashtra, Punjab, and Telangana. The organizational structure of CSA is divided into districts within the states. Within each state, CSA divides into hubs, which include one or more districts. Within the districts, there are cluster including 2-20 villages and each village has a population of 2000 farm households.

CSA supported the research efforts by identifying villages and recruiting farmers who practice natural and chemical farming methods in Kadapa, Andhra Pradesh, India. Beyond the recruited farmers, it was common for friends and family members of farmers to join them in their interviews as their lives were deeply intertwined during daily routines.

This case study was conducted through 21 semi-structured interviews over four days, from July 23, 2022, to July 30, 2022. Seventeen APCNF natural farmers and four chemical farmers were interviewed. Nine individual interviews, seven focus groups, and five family interviews took place. The focus groups typically consisted of farmers who had geographically adjacent farms. The family interviews typically had a siblings, parents, spouses, and children present. The villages visited in Kadapa District were Vemula, Bhumayyagaripalli, Velpula, Thallamapuram, Regulapalle, Thallapalle, T Velamavaripalli, Bakkanagaripalle, Gopalapalli, Sabbanagunthapalli, and Kupaalapalli. Before each day of field visits, CSA field staff in the cluster would reach out to the farmers and ask them to gather in one common location for the interviews to take place.

The interview questions were initially written in English and translated by myself into Telugu to ask the farmers during the interviews (interview questions included in appendix). Prior

to the interview, interviewees were asked for consent and informed that the primary researcher recorded the conversation. The interview began with general demographic information and then led to questions about their crop choices, challenges, and motivations for transitioning to natural farming. Overall, the interview duration was 20 minutes to 1 hour, depending on how much the farmer was willing to share about their experiences and build on the conversation guided by the pre-written interview questions. The interviews were recorded on a mobile phone and transferred daily to UTBox for secure storage. The recordings and transcripts were coded with the data (mm/dd) and farmer initials to ensure anonymity. For analysis, the Telugu recordings were transcribed into English to code for themes.

A thematic analysis of the interviews, including field notes, was done following Lochmiller's (2021) process of qualitative data analysis to answer the main research question of why farmers in Kadapa were transitioning to and sustaining natural farming practices. As I was transcribing, I took note of the repeating factors farmers brought up in regards to their transition to natural farming. I would highlight these factors in the document and then I was able to code the responses into three themes: (1) CSA and community support, (2) the impact of chemicals on farming practices, and (3) the improvement of both quantity and quality of produce. These influencing factors highlight the farmers' perspectives on natural farming. As the ground-level actors of sustainable agricultural practices, it is important to hear their voice on this topic.

All of the insights in the following section are based on observations from the field visits and select quotes are included that best exemplify each theme, as many farmers had similar experiences with natural farming, APCNF, and CSA. The quotes included in the thematic analysis section are attributed to farmers whose names are held confident with pseudonyms.

CSA and Community Support

The learning curve for natural farming is difficult for farmers because they are taking a risk on their crop with new a new farming practice. CSA supports these farmers by creating a system of knowledge transfer from the field staff to the farmers which then promotes peer learning and resource sharing among the farmers in the village.

The APCNF program is organized by the government of Andhra Pradesh, which provides the resources to train field staff to work with farmers to transition to natural farming. The frontline workers of this program are provided by Rythu Sadhikara Samstha (RySS) and CSA. These field workers help maintain Farmer Producer Organizations (FPO), market the naturally farmed produce, and advocate for fair prices for the farmers.

In my time with the field staff, I saw that the government's involvement in the program didn't seem to sway farmers to natural farming. However, the farmers did relate the APCNF program to the CSA staff with whom they interacted with on a daily basis. The field staff typically came from one of the villages in the cluster that the FPO functioned in. This local advantage gave the farmers the opportunity to connect with the fieldworkers with whom they were familiar with. The FPO board members are often model natural farmers from villages where they have successfully conducted natural farming, become certified, and have a passion for advising other farmers on their transition. The farmers are also open to learning from the field staff; as Kupalapalle farmer "Subbanna" expressed, "CSA staff has been behind me since 2017, and they consulted with me and taught me how to do it. NPM farmers in the area are close by, so I had no problem learning from people and supporting me."

NPM refers to using bio-pesticides and mechanical pest management practices, which function through understanding the pest life cycle to manage pests successfully. I observed

villages where farmers do not immediately transition to natural farming on their entire land. They often start by replacing their use of pesticides with non-pesticide management (NPM) practices. Subbanna's case is an excellent example of the community support aspect of APCNF. Subbanna received information from CSA, which gave him enough confidence to begin natural farming. His effort was sustained by the community support he is receiving from the NPM farmers in his village.

Prior to CSA's presence, it was common for farmers to have taken their own initiative, at a limited capacity, to reduce their chemical use. Several farmers interviewed had an initial interest in natural farming practices, which CSA identified and supported. "Reddamma" an organic horticulture farmer from T. Vellammavari Palle stated, "[CSA] just came. Before, I used chemicals, and I was boiling neem and spraying that, but they came and helped explain what to spray." Many farmers were already using some generic natural farming practices, such as using manure to improve soil fertility and variations of neem concoctions as a natural pesticide. Neem is a native tree to India and is abundantly found in the Kadapa region, where it has been used as a pesticide against sucking pests and mealy bugs for generations. Some farmers had a baseline knowledge of natural pest management, which was supplemented by CSA's support. For example, Reddamma was boiling neem and using it as a pesticide, but with CSA's guidance, she learned how to make neemastra, a fermented mixture of neem, cow urine, and cow dung, which increases the effectiveness of the natural pesticide. This type of support to the farmers helps them transition to natural farming as it is built on their existing interest and knowledge.

The APCNF program in Kadapa empowers farmers to explore natural farming with confidence and minimal risk, potentially attributed to the strong community support and expert guidance from CSA. By placing farmers at the center of the program, it fosters a grassroots

movement that arises from within the community rather than imposing change from outside. As a result, farmers are able to tailor their approach to natural farming to their unique circumstances, ensuring greater feasibility and success.

Impact of Chemicals

Conventional farming has traditionally relied on synthetic agrochemical, but rising prices of these chemicals in recent years have prompted many farmers to reconsider their practices. The farmers are exploring natural alternatives to chemical pesticides and fertilizer, not only to reduce costs but also to mitigate the long-term impacts of chemical exposure on themselves, their crop, and their land. In Kadapa, this growing concern for health and sustainability has lead more and more farmers to make the transition to natural farming.

Price of Chemicals

In Andhra Pradesh and across India, the standard practice of farming involves the use of agrochemicals such as urea, phosphate, and nitrogen. However, the prices of these chemicals fluctuate and impact the farmers who have become dependent on them to support their crops. During the period of the interviews, the prices of these chemicals had heavily increased due to supply chain issues. Although the farmers did not delve extensively into this point during their interview, it was notable that they raised the issue of increased chemical costs multiple times across all the interviews. They related this increase in price and compared it to the reduced expenses they experienced with natural farming. While the price increase wasn't a primary factor that led farmers to turn to natural farming, it was one of the reasons why farmers began showing more interest in natural farming practices. Some farmers also expressed that the effectiveness of the chemicals had decreased, and farmers were finding that their crops were being ruined by the use of these chemicals.

Crop Health

Farmers in my interviews recollected chemical-free farming from their grandparents' generation. However, by the time they made farming their livelihood, they had become dependent on chemicals. Some farmers even noticed that their crops were withering due to the prolonged use of chemicals. "Rajanna," an experienced banana farmer in Vempalle, recollected that "before organic manures, the trees were dying because of the chemicals used, and there were more withering of trees." He also spoke in depth that he began natural farming because it was healthier for the crop and himself. He shared with me multiple times that he was much happier with natural farming because it allows him to practice integrated farming with the animals he already had on the farm. Conventional farming practices strip the soil of nutrients which would have been the cause of the withering of the trees, and natural farming works to revitalize the soil so the increase nutrient content could have been the cause of the difference Rajanna observed.

Recently there has been a revitalized interest in natural farming to save the crop and nurture the land, and farmers are reclaiming traditional farming practices through natural farming. Many farmers have transitioned to natural farming in the last ten years, either by fully committing to natural farming on plots of their land or incrementally converting their fields. Some farmers begin natural farming by replacing their chemical remedies/inputs with natural options slowly, such as replacing fertilizers with cow manure and pesticides with neem concoctions. As they gain confidence in the strength of the remedy, they begin to use it more at a larger scale.

The Relation Between Crop Health and Human Health

The natural farmers I interviewed addressed personal health benefits from the perspective of crop health. In recent years, farmers have begun noticing the physical health implications of

chemical use. “Abdul,” a groundnut farmer from Regullapalle, stated, “you get various diseases from spraying chemicals and from consuming food with it on. So, when we wait and put green grass on top and we had *chenum* and after that we put phosphate and urea and after that we put fertilizers '*eruvu*'. In these cases, only the *eruvu* is working. Urea, phosphate, and potassium are not working. The diseases are going down.” Abdul had an issue with '*chenum*', which is a pest that typically impacts crops in the Kadapa region. I asked the him to elaborate on that last part about diseases, and he linked both aspects of personal and crop health together stating that not only do illnesses in humans decrease due to reduced use of chemicals but “diseases in plants too. And it's healthy when [everyone] eats that harvest too”.

Beyond crop health and personal health concerns due to chemical use, other farmers also expressed that they were satisfied with their transition to natural farming because they would be able to provide nutritious food for themselves. I initially went to Kadapa with the preconceived notion that natural farming took more time and effort, which might be a deterrent for farmers to transition to natural farming. However, when comparing the two practices, “Vijay,” a tenant farmer from Nakaldi Anantapuram, reflected on the time used for each practice and stated that “[natural farming] feels normal. It's nutritional food, so even though it's more effort, it doesn't feel like hard work.” This farmer's distinction between effort and hard work shows that he is satisfied with the amount of work he is putting into the natural farming practices he implemented on his field. The concept of improved nutrition is built into the APCNF model as it promotes the use of inter-crops and border crops, which are typically the native crops of millet and lentils. The goal of these crops is to provide the farmer with nutritious food that they can take home and sell any excess.

The adverse effects of agrochemicals on crop yields and human health have led many farmers to seek a transition to natural farming. Moreover, the rising cost of chemicals has prompted farmers to adopt a more sustainable and cost-effective approach to farming. This shift towards natural farming not only addresses the economic stress factors of agrarian distress but also promotes a holistic approach to farming that supporting environmental sustainability and human well-being.

Improved Quantity and Quality of Produce

In order for natural farming to truly be a sustainable farming practice, it is imperative that farmers experience an increase in both the quantity and quality of their yield, which should ideally translate into higher profits based on fair market prices. However there are extraneous factors such as climate conditions which impact crop success. Duddigan et al. (2022) conducted a study in Andhra Pradesh comparing the effectiveness of chemical, conventional, and natural farming practices across different districts. The results of this study, which are discussed in detail under the sustainable agriculture section in this paper, indicated that the Zero Budget Natural Farming (ZBNF) program's effectiveness varied depending on the district. In summary, districts with ample rainfall showed lower yields from ZBNF practices, while arid districts in the south saw increased output. However, in northern districts, increased precipitation had a negative impact on ZBNF yields. Therefore, it is crucial to consider the geographical context of Kadapa to understand the responses of the farmers.

Kadapa, an arid district in southern Andhra Pradesh (Figure 1 and 2), has been more successful in ZBNF practices, and many farmers that were interviewed reported an increase in their yield. For instance, “Indira”, a female groundnut farmer from Sabanagunta Palle, noticed

that her groundnut harvest increased from 12 bags to 40 bags once she shifted to natural farming. While some farmers did not notice a yield increase, they did observe that their soil had softened, which they believed was an outcome of the natural farming practices. Which many farmers and CSA staff members pointed out as we walked in the field as *whapsa* soil. This soil was typically soft and was a contrast to the dry, sandy and rocky soil of the region. Additionally, many farmers shared that the quality of their produce had visually improved, two farmers particularly described their produce as ‘shiny.’

“Rajanna,” a Vempalle farmer, stated that, "When I eat [the produce] for myself, I give some to my friends. And when I give it to them, they say it looks really special, it shines. And I say, what do I know, and then after they eat it, I say, mine is 100% organic and no spray medicines. I only spray with leaves and scraps, no chemicals. And they say, ‘oh aha, so like the old practices’." The farmer's testimony highlights the interplay of factors that farmers value. This particular farmer took pride in his transition and the produce he was growing through natural farming practice. I am not fully aware if this farmer is a certified organic farmer but many times farmers did not differentiate between organic and natural farming practices and relate them to certifications. The farmers related these sustainable farming practices to reduced chemical use that improves their land and crop.

The idea of “shining” produce also came up in conversation with “Kumar Rao” from T.Vellammavari Palle who stated that, “for papaya there more viruses so I use this mix with 5 types of leaves to spray on the plant. Same mix for papaya, vonkaya (eggplant), tomato. Then the last time CSA sir came he told me about some 3 oils. One is kayanune, vepanune, one other one.” When I asked about the purpose of the oils he responded that it was “for the plants, for the fruit to have shining. The vonkaya is shining.” The accompanying CSA staff member and Kumar

Rao helped explain that the oils help protect the fruit and make it more appealing to consumers. The conversation with Kumar Rao also reinforced the importance of CSA and community support while also highlighting the central theme of the quality of the produce. It was evident that these farmers were committed to enhancing the appeal of their products to consumers through the use of natural farming techniques.

The positive outcomes of natural farming practices, the APCNF program, and CSA support in Kadapa are reflected in the increased quantity and quality of produce. However, the potential benefits of these improvements in terms of higher returns for farmers are subject to seasonal demand and market prices for their crops. Nonetheless, overall, the farmers express a positive outlook on the results of natural farming and seem satisfied with the outcomes once they become confident in these practices.

Through fieldwork in Kadapa, I gained a deeper understanding of the brevity of agrarian distress, surpassing what I had previously read in literature. As I worked with farmers, it became apparent that the ecologic, economic, and social stress factors discussed in the agrarian distress section were present. Additionally, I observed the impact of ZBNF and APCNF on the farming community. It was inspiring to see CSA working alongside small and female farmers, providing education on natural farming practices and offering support as they progressed. While measuring the full extent of agrarian distress is challenging, CSA's efforts targeting the main stress factors and empowering farmers are making a significant difference.

Discussion

In Kadapa, AP, I observed the ecological, economic, and social stress factors outlined in the agrarian distress section. Working closely with various field staff teams, I witnessed the

important relationships they maintained with the communities they served. Through interviews with farmers, I aimed to identify the factors that motivated them to transition from chemical farming to natural farming and sustain their practices.

Farmers are facing challenges from changing growing seasons and unpredictable weather events, making it difficult to recover from setbacks year after year. The three factors identified above are interconnected and can help reduce uncertainty in the community. Natural farming is seen as a way to recover the soil, leading to better yields and increased profits. CSA and community support can help farmers learn and execute natural farming techniques.

While farmers have an inherent connection to nurturing their land, many do not want their children to farm. In my fieldwork, they expressed pride in their work and wanted to teach their children the value of nurturing soil and crops, but they also recognized that a lack of education or funds has led them to farm. Despite their desire for their children to pursue higher education and careers beyond farming, they continue to work hard to provide for their families.

The older children of these farmers see the value in living in a village community and environment but question whether farming itself is sustainable. They face challenges such as working in the blistering sun, unpredictable weather, and the physical toll of chemical use on their bodies. All the farmers I spoke with had children attending school, and they expressed their desire for their children to receive higher education and pursue careers beyond farming. The shift to natural or organic farming comes from concerns about health, increased profits, and a general interest in sustainable agriculture.

Ultimately, the true motivation for these farmers is to find a better future for their families. However, a deeper discussion on ecological sustainability and holistic sustainable livelihoods that empower farmers and farming communities is needed. Programs supporting

sustainable agriculture must promote a mission centered around creating sustainable livelihoods for farmers.

Conclusion

The APCNF and ZBNF programs aim to address the multifaceted issues underlying agrarian distress by prioritizing the success of small and marginal farmers. These farmers face numerous uncertainties that contribute to crop loss. However, the programs are making significant strides in combating these challenges through sustainable agriculture practices. While the literature provides a wealth of knowledge, the voice of the farmers is missing in this movement toward sustainability. This paper seeks to assess the efficacy of the programs and encourage the adoption of sustainable agriculture practices throughout India. To achieve this purpose, the discussion on ecological, economic, and social stress factors of agrarian distress is presented in parallel with the goal of sustainable agriculture programs in India, specifically the APCNF program in Andhra Pradesh.

The Kadapa, Andhra Pradesh case study delves into the perspectives of farmers as they transition towards and continue natural farming, aiming to understand their efforts and outcomes. Through conversations with the farmers, three recurring themes emerged:

1. The importance of CSA and community support
2. The negative impact of chemicals on farming practices
3. The improvement in both the quantity and quality of produce

By organizing the ideas behind these themes chronologically, it becomes clear how Kadapa farmers gained confidence in sustainable agriculture practices and how community support was crucial in facilitating peer learning. Initially, farmers used chemicals but were also

knowledgeable in basic natural farming practices. However, the presence of CSA and enthusiastic community support in the area provided the necessary resources for learning about effective natural farming practices. As a result of this support, farmers began experimenting with natural farming on a portion of their land. They observed that low-cost natural alternatives to chemical pesticides and fertilizers were efficient and improved crop health and soil fertility. With continued support from CSA and the community, farmers expanded their natural farming practices to more areas of their land. This general journey and the themes emerged as essential elements for a successful transition towards natural farming, as highlighted in my case study.

To gain a more comprehensive understanding of the success of natural farming, future research must take a holistic approach. One possible study could be a longitudinal cohort analysis comparing populations that have transitioned to natural farming with a control group that uses conventional farming practices. This type of research could provide insight into the long-term benefits of natural farming and the overall effectiveness of the APCNF program. Moreover, studying how farmers can best be educated on environmental literacy and sustainability is crucial. This may require a parallel study to explore how sustainable agriculture programs can be effectively taught in collaboration with rural communities.

This paper highlights the preliminary evidence supporting the advantages of natural and sustainable agriculture for addressing agrarian distress. The next crucial step is to understand better how to promote more sustainable agriculture practices across India and globally to ensure better sustainable food systems from farmers to consumers.

Limitations

The primary consideration for sustainable agriculture is that there is no blanket solution to agrarian distress; each community facing agrarian distress and region under ecological stress

needs to be evaluated separately to find the proper solution. The limitation of the study conducted in Kadapa is that there is a possibility the themes found are only applicable to the district as it faces similar climate conditions and the farmers cultivate similar crops. Factors such as crops, community support, climate, culture, and politics vary from region to region, so it is not easy to replicate the same results experienced in AP in other states. However, the key takeaway is that sustainable agriculture has been shown to restore soil fertility and build trust in farmers to adopt new sustainable farming practices through community support.

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Appendix: Semi-Structured Interview Questions

Consent:

- 1) Would you like to participate in this study?
- 2) If yes, do I have permission to record our conversation?

General:

- 3) What is your age, and can you tell me how long you have been a farmer?
- 4) How much land do you have (size of landholding)?
- 5) What does your day look like during the Kharif (Rainy)/Rabi (Winter) season?
 - a) How much time do you spend on your farm every day (each season)? What do you personally do on the farm?
 - b) How do you feel about the amount of time you spend in the fields?
 - c) If you could, would you spend more time in the field?
- 6) What crops do you grow in which season?
 - a) Do you make the decision on which crops to grow? If not, then who does?
 - b) Which crops in Kharif? Which in Rabi?
 - c) Why do you grow these crops?
 - d) Have you always grown these crops? What did you grow before?
 - e) Do they intercrop in the same plot or grow different crops in different plots? Do they use border crops or trap crops for pest control?
- 7) Do you hire laborers to work on your land? Where do these laborers come from? Do the same workers work with you every year? During which season do you hire workers? Do the same workers work with you every year? Do you have any trouble with finding workers when you need them?
- 8) Do you work on other farms? If yes, do you think there is sufficient work? Do you have trouble finding work when you want to?
- 9) What kind of inputs do you use on your farm?
 - a) Do you use only chemical fertilizers and pesticides? Do you use organic inputs like compost, jeevamrutam, and kashayams? Do you use both?
 - b) Do you consider yourself a ZBNF farmer?
- 10) Who manages your land holdings?

Conclusion

- 11) Do you think your land has value? Can you describe your answer?
- 12) Do you want to keep farming?
- 13) Do you want your children to keep farming?

Author Biography

Sai Cheruvu is a recent graduate of the University of Texas at Austin, where they earned a Bachelor of Science degree in Public Health. Growing up with a family history of generational farming in Saguturu, Andhra Pradesh, India, they developed a strong interest in sustainable agriculture and its potential to address food security and environmental challenges.

During their undergraduate studies, Sai conducted research on sustainable agriculture and collaborated with the Center for Sustainable Agriculture to conduct a case study on farming practices in Kadapa, India. This experience solidified their passion for working in the field of agriculture and sustainability, leading them to pursue further education.

Currently, Sai is enrolled in the University of Michigan's Master of Science in Environment and Sustainability program, with a focus on geospatial data science. She is excited to continue their research in sustainable agriculture and explore ways to leverage technology and data to promote more efficient and environmentally-friendly farming practices.